5. Develop an Android application implementing Fragment-based UI navigation between multiple screens.

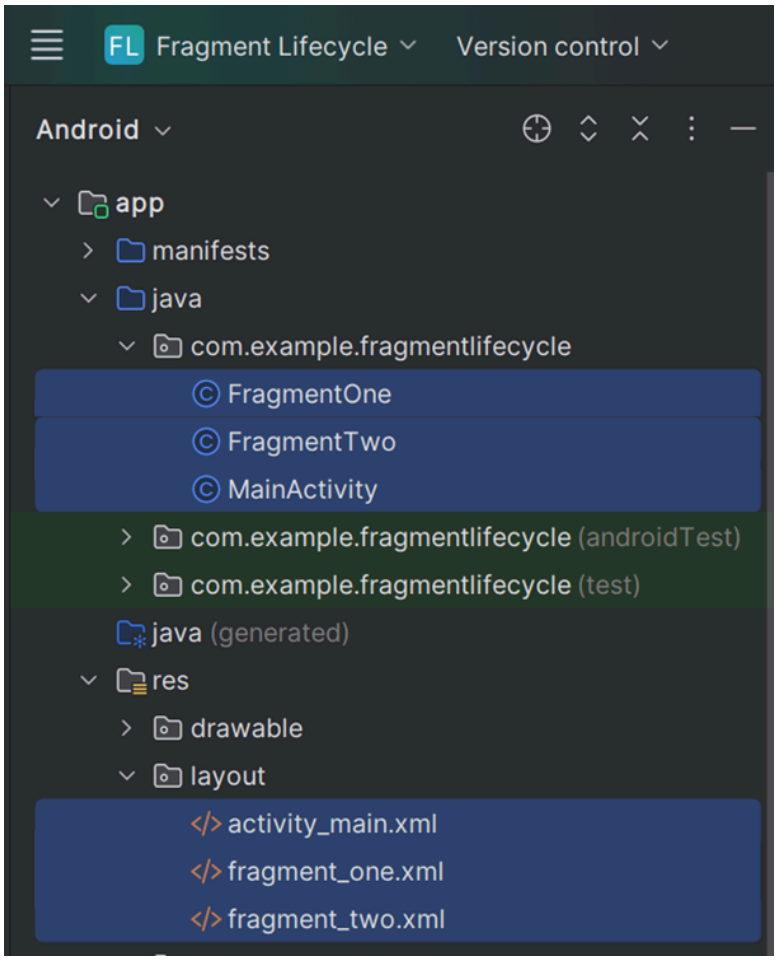
Activity Life Cycle, Activity methods, Multiple Activities, Life Cycle of fragments and multiple fragments.

Project Name: Fragment Lifecycle

Template: Empty Views Activity

Language: Java

Project Architecture: Total 6 files



I Filename: MainActivity.java

```
package com.example.fragmentlifecycle;

 import androidx.appcompat.app.AppCompatActivity;
 import android.os.Bundle;
 import android.widget.Button;

 public class MainActivity extends AppCompatActivity {

      private boolean isFragmentOneLoaded = true;
      private final androidx.fragment.app.FragmentManager fragmentManager =
getSupportFragmentManager();

      @Override
      protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_main);

        Button btnChange = findViewById(R.id.btn_change);  // Ensure ID
matches activity_main.xml
      loadFragmentOne();

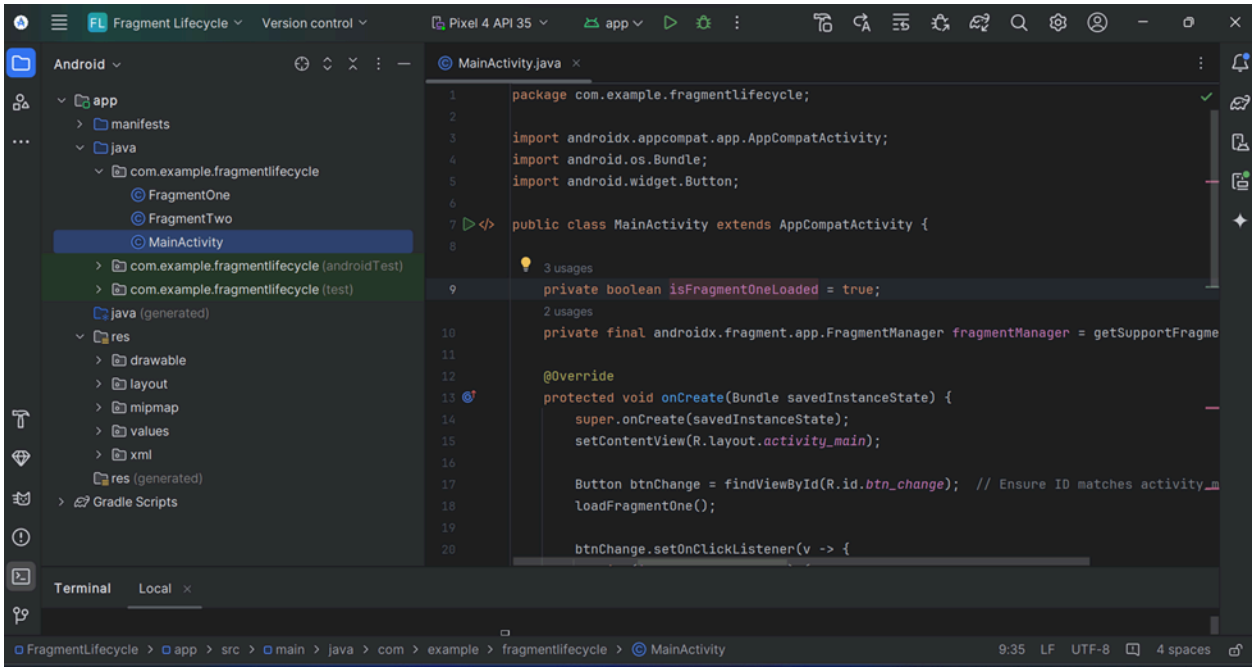
      btnChange.setOnClickListener(v -> {
            if (isFragmentOneLoaded) {
                  loadFragmentTwo();
            } else {
                  loadFragmentOne();
            }
      });
      }

      private void loadFragmentOne() {
         androidx.fragment.app.FragmentTransaction transaction =
fragmentManager.beginTransaction();
      FragmentOne fragment = new FragmentOne();
      transaction.replace(R.id.fragment_holder, fragment);  // Ensure ID
matches activity_main.xml
      transaction.addToBackStack(null);
      transaction.commit();
      isFragmentOneLoaded = true;
      }

      private void loadFragmentTwo() {
         androidx.fragment.app.FragmentTransaction transaction =
fragmentManager.beginTransaction();
      FragmentTwo fragment = new FragmentTwo();
      transaction.replace(R.id.fragment_holder, fragment);  // Ensure ID
matches activity_main.xml
```

```
        transaction.addToBackStack(null);
        transaction.commit();
        isFragmentOneLoaded = false;
        }
}
```

Path: app>java>com.example.fragmentlifecycle>MainActivity.java



II

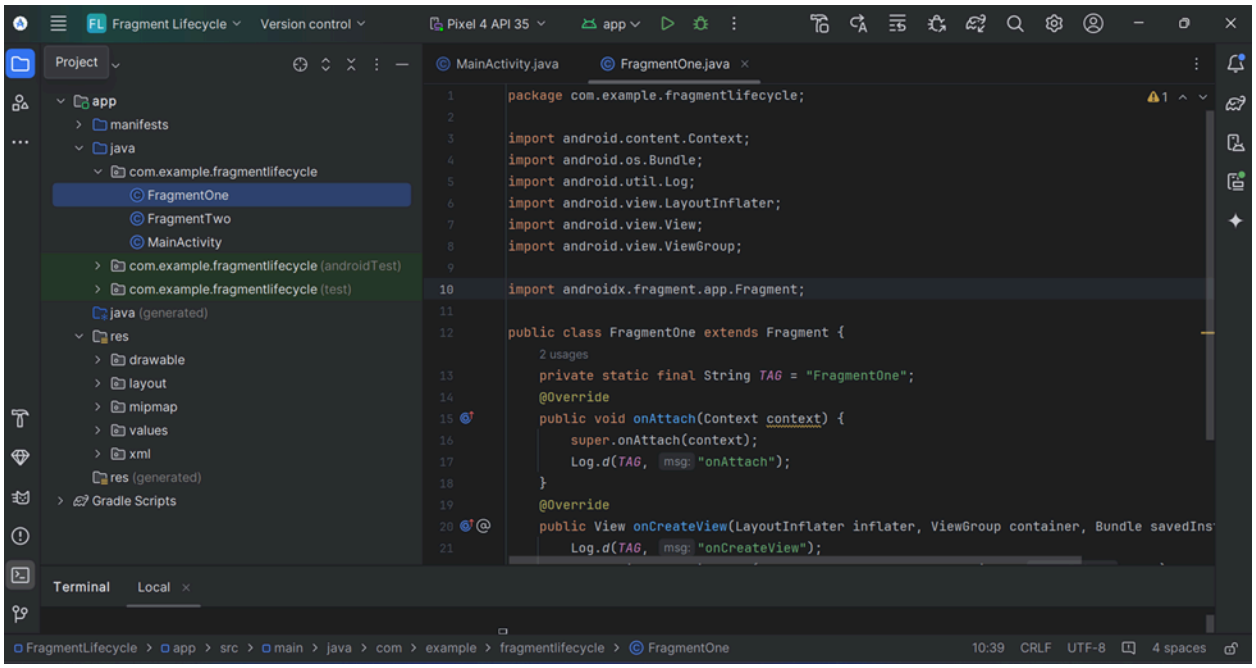
Filename: FragmentOne.java

```
package com.example.fragmentlifecycle;
import android.content.Context;
import android.os.Bundle;
import android.util.Log;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import androidx.fragment.app.Fragment;
public class FragmentOne extends Fragment {
        private static final String TAG = "FragmentOne";
        @Override
        public void onAttach(Context context) {
        super.onAttach(context);
```

```
        Log.d(TAG, "onAttach");
        }
        @Override
        public View onCreateView(LayoutInflater inflater, ViewGroup container, Bundle
savedInstanceState) {
        Log.d(TAG, "onCreateView");
        return inflater.inflate(R.layout.fragment_one, container, false);
        }

}
```

Path: app>java>com.example.fragmentlifecycle > FragmentOne.java



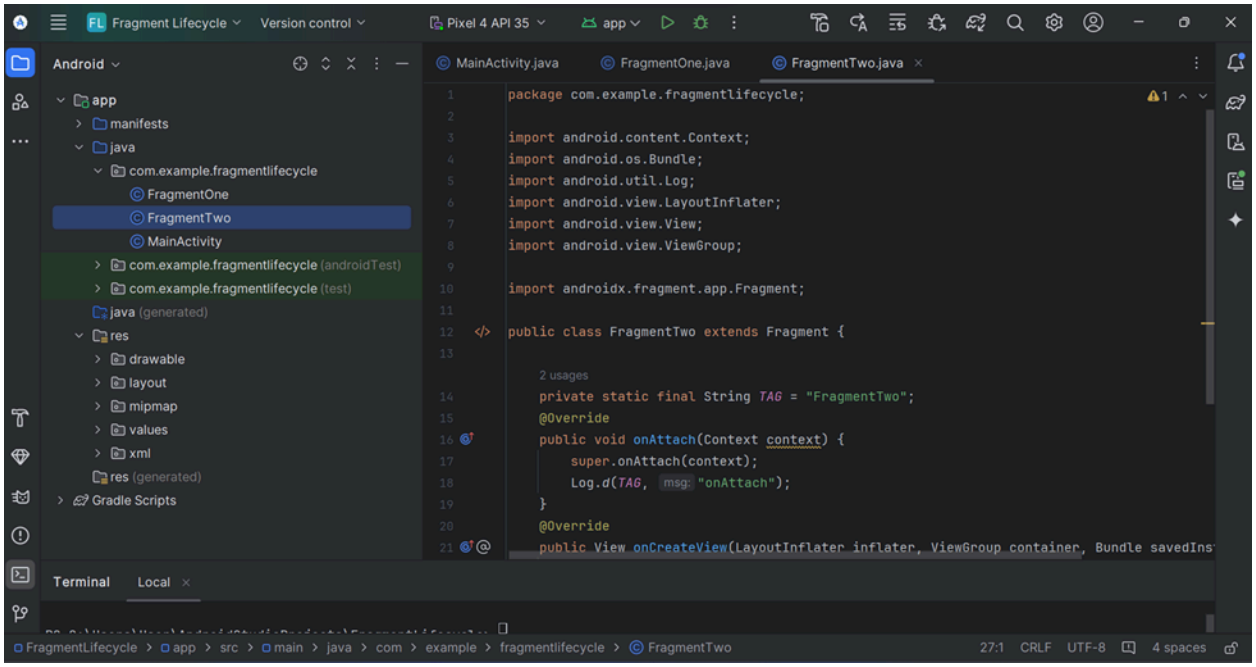
III

Filename: FragmentTwo.java

```
package com.example.fragmentlifecycle;
import android.content.Context;
import android.os.Bundle;
import android.util.Log;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import androidx.fragment.app.Fragment;
public class FragmentTwo extends Fragment {
        private static final String TAG = "FragmentTwo";
```

```
        @Override
        public void onAttach(Context context) {
        super.onAttach(context);
      Log.d(TAG, "onAttach");
        }\
        @Override
        public View onCreateView(LayoutInflater inflater, ViewGroup container, Bundle
savedInstanceState) {
        Log.d(TAG, "onCreateView");
        return inflater.inflate(R.layout.fragment_two, container, false);
        }
}
```

Path: app>java>com.example.fragmentlifecycle > FragmentTwo.java



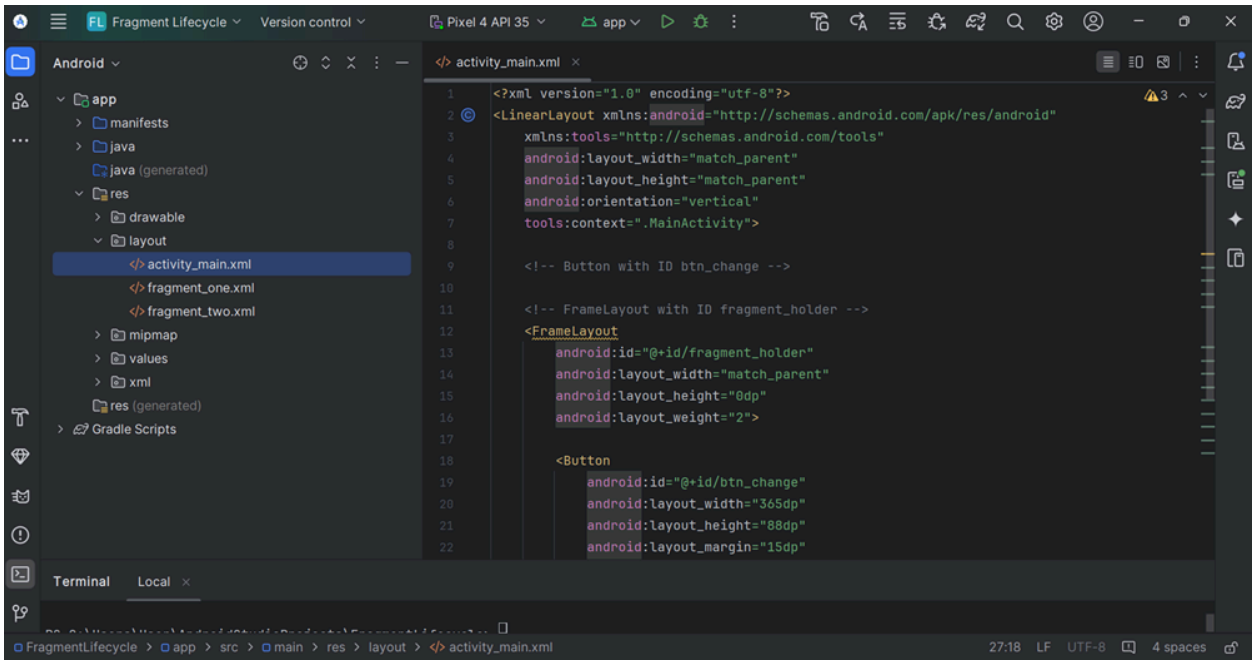
IV

Filename: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
        xmlns:tools="http://schemas.android.com/tools"
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:orientation="vertical"
        tools:context=".MainActivity">
```

```
        <!-- Button with ID btn_change -->
        <!-- FrameLayout with ID fragment_holder -->
        <FrameLayout
        android:id="@+id/fragment_holder"
        android:layout_width="match_parent"
        android:layout_height="0dp"
        android:layout_weight="2">
        <Button
        android:id="@+id/btn_change"
        android:layout_width="365dp"
        android:layout_height="88dp"
        android:layout_margin="15dp"
        android:layout_marginTop="15dp"
        android:backgroundTint="@android:color/holo_green_dark"
        android:text="Change Fragment"
        android:textAllCaps="false"
        android:textSize="24sp" />
        </FrameLayout>
</LinearLayout>
```

Path: app>res>layout>activity_main.xml

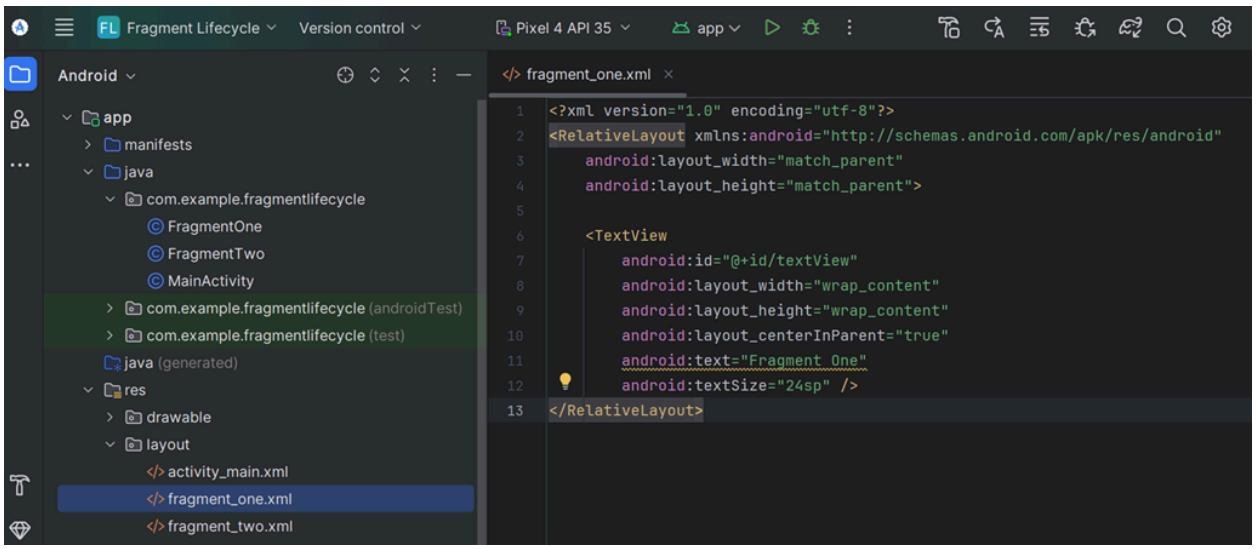


V

Filename: fragment_one.xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
        android:layout_width="match_parent"
        android:layout_height="match_parent">
        <TextView
        android:id="@+id/textView"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_centerInParent="true"
        android:text="Fragment One"
        android:textSize="24sp" />
</RelativeLayout>
```

Path: app>res>layout>fragment_one.xml



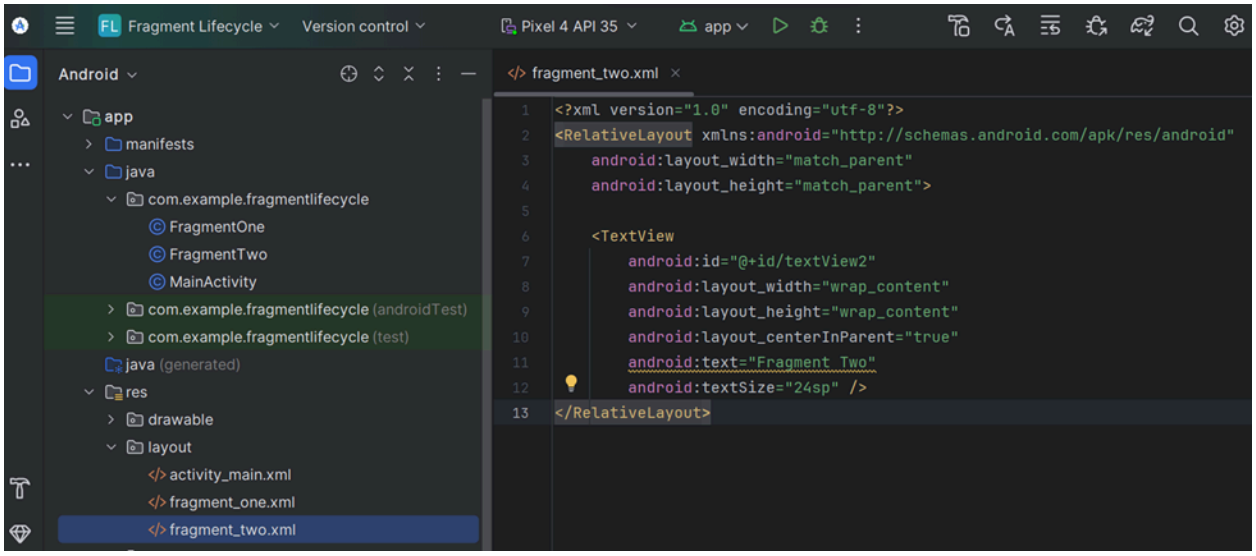
VI

Filename: fragment_two.xml

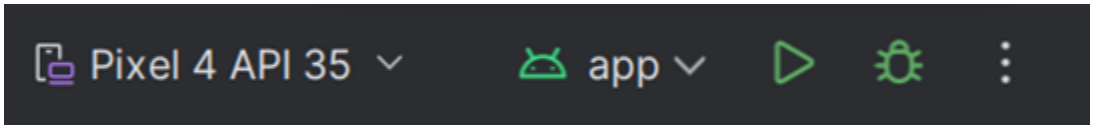
```
<?xml version="1.0" encoding="utf-8"?>
```

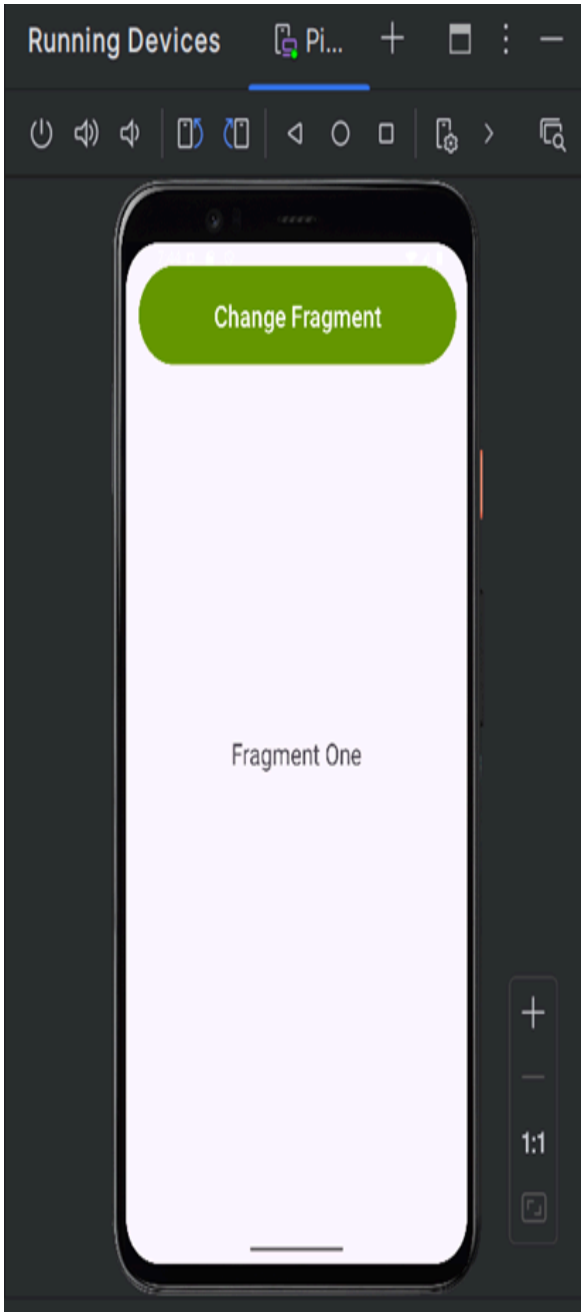
```
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
        android:layout_width="match_parent"
        android:layout_height="match_parent">
        <TextView
        android:id="@+id/textView2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_centerInParent="true"
        android:text="Fragment Two"
        android:textSize="24sp" />
</RelativeLayout>
```

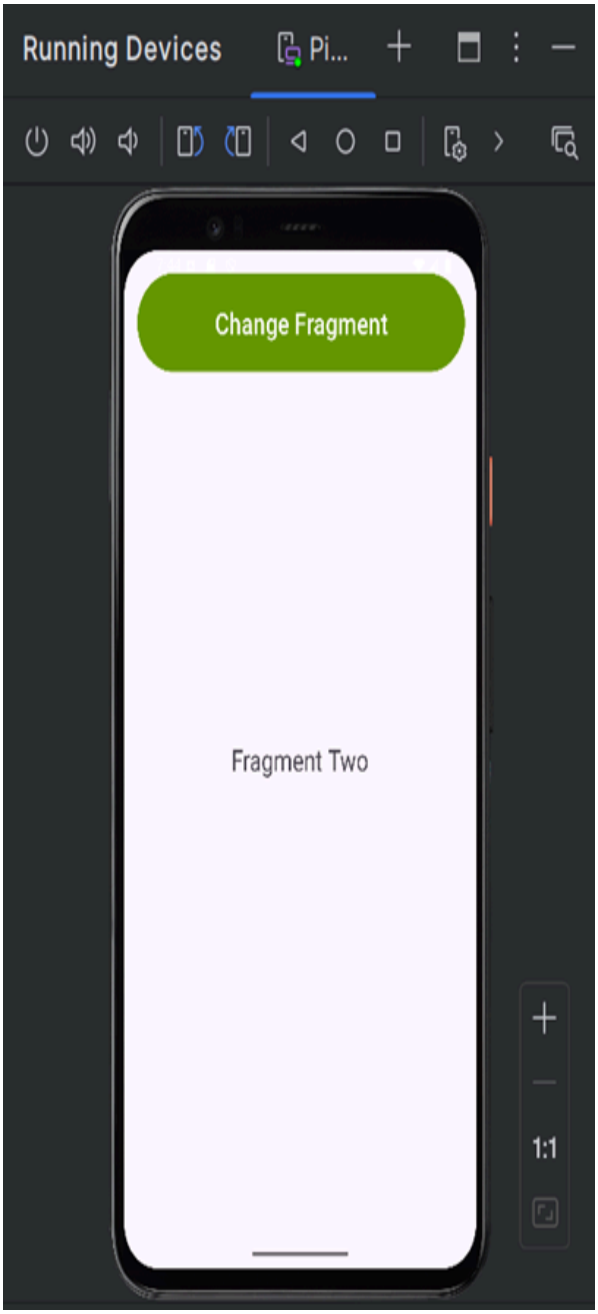
Path: app>res>layout>fragment_two.xml



Above the Android Studio near Main Menu Click the Play button to Run the Program




Running Devices
Pi...
Change Fragment
Fragment One
1:1


Running Devices
Pi...
Change Fragment
Fragment Two
1:1